

Federated Learning TP1

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Course: AI for Connected Industries

Module: Distributed and Federated Learning

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Abstract—This report presents the implementation, and evaluation of TP1 from the Federated Learning course lab. a federated learning training and evaluation pipeline was developed using the Flower framework, with Fashion-MNIST serving as the dataset for training and testing. Client-side data heterogeneity was introduced and controlled through Dirichlet partitioning to simulate realistic non-IID conditions. A simple FedAvg model was implemented using neural networks built in PyTorch. Dedicated client and server applications were created to orchestrate the federated training rounds, and log system output. Configurations based on heterogeneity, clients, and hyperparameters were simulated in controlled setting, the results show that data heterogeneity (α) and sampling ratios are dominant factors influencing convergence in federated setup.

Index Terms—Federated Learning, Flower, non-IID data, alpha Dirichlet, client sampling, FedAvg, FedSGD

I. INTRODUCTION

Federated learning (FL) enables a global model to be trained across many clients without sharing raw data. Each client trains locally on its own partition and sends only model updates to a central server, that aggregates them into an improved global model. Key challenges in implementation are heterogeneity of client data, communication cost, and client counts. These are explored under varied experimental conditions.

II. REPORT STRUCTURE

The end to end orchestration and experimentation of federated learning is hosted on GitHub for replicability. (Git Repo: dfl-tp-workspace). Report covers the results from different experiments on flower where various hyperparameter values are tested in combination. The plots from every tp experiments are recorded in a single pdf added as annexure (annexure-output-plots.pdf).

III. BACKGROUND

A. Federated Averaging

FedAvg is the standard baseline algorithm for federated learning. The server aggregation is carried using the following method:

$$w^{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{t+1} \quad (1)$$

where $n_k = |\mathcal{D}_k|$ is the number of training samples on client k and $n = \sum_k n_k$ is the total. No raw data leaves any client at any point.

B. Dirichlet Partitioning

Data heterogeneity is controlled using Dirichlet distribution $\text{Dir}(\alpha)$. A small α (e.g. $\alpha = 0.01$) produces highly skewed partitions where each client holds samples from very few classes. A large α (e.g. $\alpha = 100$) produces near-uniform partitions that approximate independent and identically distributed (IID) data. This parameterization allows for systematic control of heterogeneity across experiments.

IV. EXPERIMENTS

TABLE I: Default Setup: Fixed and varied parameters for all experiments.

Parameter	Value
Dataset	FashionMNIST (60 000 train, 10 000 test)
Model	CNN (2 conv layers + 2 FC layers)
Learning rate	0.01
Batch size	64
Default alpha	0.1
Default Fraction train	1
Default Clients	50
Varied alpha value	$\alpha \in \{0.01, 5, 100\}$
Algorithm	{FedAvg+SGD, FedAvg+ADAM, FedSGD}
Clients n	{5, 20, 50}
Fraction train	{0.1, 0.5, 1}
Local epochs E	{1, 5}
Communication rounds	{10, 20}

A. Effect of data heterogeneity (α) on FedAvg Convergence

1) Expected Behavior

FedAvg convergence should improve monotonically with α . Near-IID data ($\alpha = 100$) should yield the fastest convergence and the highest final accuracy because client gradients point in similar directions, unlike non-IID data ($\alpha = 0.01$).

2) Observation and Deviations

- $\alpha = 5$ achieves the highest final central accuracy despite representing only moderate heterogeneity (refer Table II).
- $\alpha = 0.01$ achieves the highest train accuracy despite having the lowest central accuracy, revealing severe over-fitting gaps. (refer Figure 2)
- Aggregation noise at large n .** Under strict IID ($\alpha=100$) with $n=50$ clients, each client partition is a near-identical random subset, producing low-variance but redundant updates. Moderate heterogeneity ($\alpha=5$) introduces complementary class coverage across clients, yielding more informative aggregated gradients per round. (refer Figure 1)

TABLE II: Final round metrics for each α setting.

Alpha	Train acc	Train loss	Central acc	Central loss
$\alpha = 0.01$	97.2%	0.100	51.9%	1.665
$\alpha = 5$	75.0%	0.661	66.6%	1.470
$\alpha = 100$	73.0%	0.693	61.1%	1.660

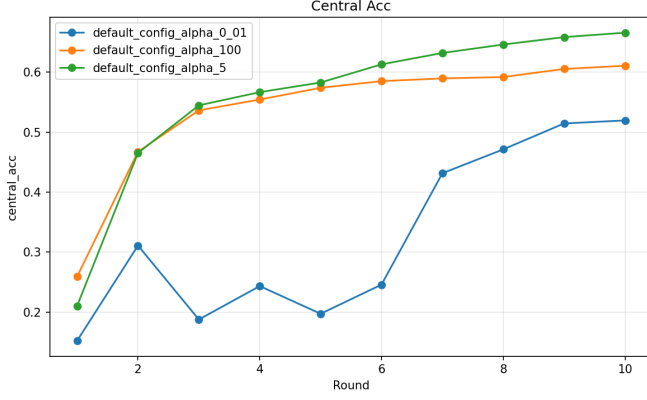


Fig. 1: Central test accuracy for $\alpha \in \{0.01, 5, 100\}$ with 50 clients

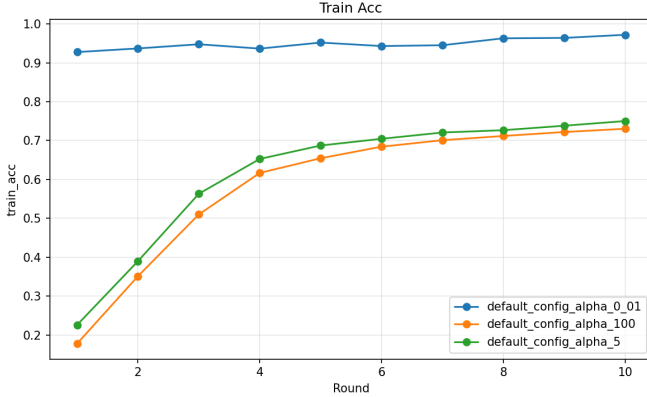


Fig. 2: training accuracy for $\alpha \in \{0.01, 5, 100\}$ with 50 clients

B. Experiment: Effect of Number of Clients n

1) Expected Behavior

Larger client pool should improve the global model as more clients cover a greater diversity of the data distribution. With $\alpha = 0.01$, each client data is dominated by approximately one class. Having more clients therefore means more classes are represented per aggregation round, which should yield faster convergence and higher final accuracy.

2) Observation and Deviations

- $n=20$ achieves the highest final central accuracy followed by $n=5$ and $n=50$. More clients do not improve convergence under severe non-IID data. (refer Table III)
- With 50 clients the model is suffering from conflicting steps from clients, hence leading to greater volatility. Due to the higher heterogeneity at $\alpha = 0.1$, client updates

tend to contradict leading to slower convergence. (refer Figure 3)

- Large spikes in federal eval loss with $n=50$ suggests conflicting updates by clients due to non-IID conditions. (refer Figure 4)

TABLE III: Final round metrics for each client count at $\alpha = 0.1$.

Clients	Train acc	Train loss	Central acc	Central loss
$n = 5$	95.8%	0.145	64.3%	1.462
$n = 20$	94.8%	0.155	69.0%	0.695
$n = 50$	90.9%	0.276	40.3%	2.255

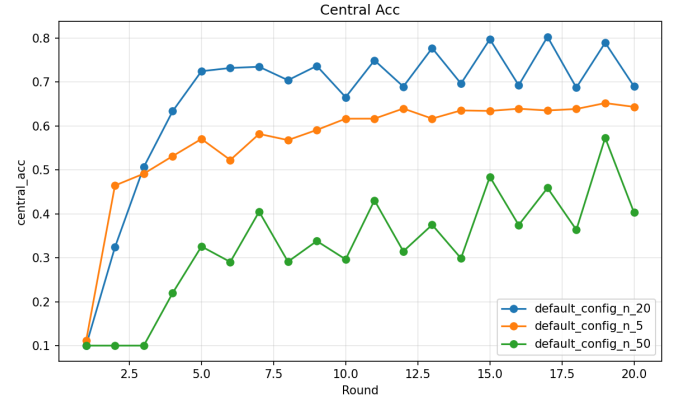


Fig. 3: Central test accuracy for client $n \in \{5, 20, 50\}$

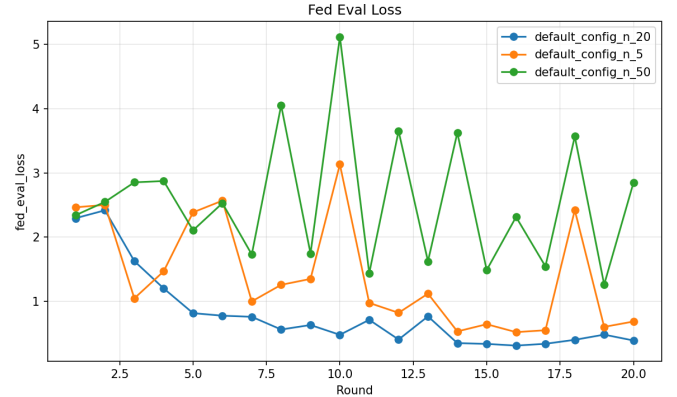


Fig. 4: federal eval loss for client $n \in \{5, 20, 50\}$

C. Experiment: Effect of Sampling Ratio f

1) Expected Behaviour

Higher sampling ratio means more clients contribute per round, providing broader coverage of the data distribution and a more reliable weighted average in Equation 1. Full participation should converge fastest and achieve the highest accuracy, while $f=0.1$ (only 5 clients per round) should be the slowest and most noisy.

2) Observation and Deviations

- **Why $f=1.0$ underperforms $f=0.5$.** This result shares the same root cause as the $n=50$ collapse observed in the client count experiment Section C). With all 50 clients participating, each client's contribution is weighted by n_k/n with $n=50$, heavily diluting individual updates under $\alpha=0.1$ conditions. (refer Table IV)
- **Why $f=0.1$ fails.** With only 5 clients sampled per round from a pool of 50 under $\alpha=0.1$, each round receives gradient signal from at most 3-5 classes out of 10. The aggregated model is updated strongly in favour of those classes while the remaining classes receive no correction. (refer Figure 5)

TABLE IV: Final metrics for each sampling ratio with $\alpha=0.1$.

Sampling	Train acc	Train loss	Central acc	Central loss
$f = 0.1$	75.3%	0.784	41.6%	3.578
$f = 0.5$	88.6%	0.390	52.9%	1.270
$f = 1.0$	90.9%	0.276	40.3%	2.255

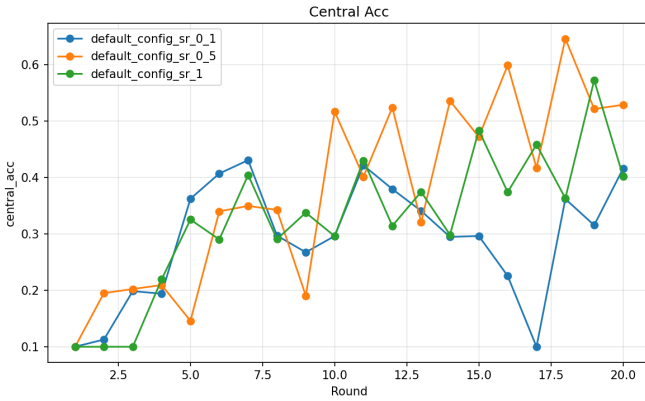


Fig. 5: Central test accuracy for sampling rate $\in \{0.1, 0.5, 1\}$

D. Experiment: Effect of different local epochs

1) Expected Behavior

More local epochs should help near-IID data by extracting more signal per round. More local epochs should hamper heterogeneous data training by amplifying client drift. $\alpha=100$ tend to outperform $\alpha=0.01$ since aggregation is more stable.

2) Observation and Deviations

- Train loss for $\alpha=100$ and $L=5$ decreases steadily and converges with low variance, consistent with near-IID gradient alignment.
- With near IID more local epochs allows model to converge faster as gradients across clients are similar. (refer Figure 6)
- As the data become heterogeneous, higher value of local epochs can introduce client drift.

TABLE V: Final round metrics for each local epoch 1 and 5.

epochs	Train acc	Train loss	Central acc	Central loss
$\alpha=0.01, l=1$	98.6%	0.04	64.1%	0.99
$\alpha=0.01, l=5$	99.7%	0.01	64.9%	0.85
$\alpha=100, l=1$	94.8%	0.15	78.2%	0.56
$\alpha=100, l=5$	98.7%	0.39	84.7%	0.38

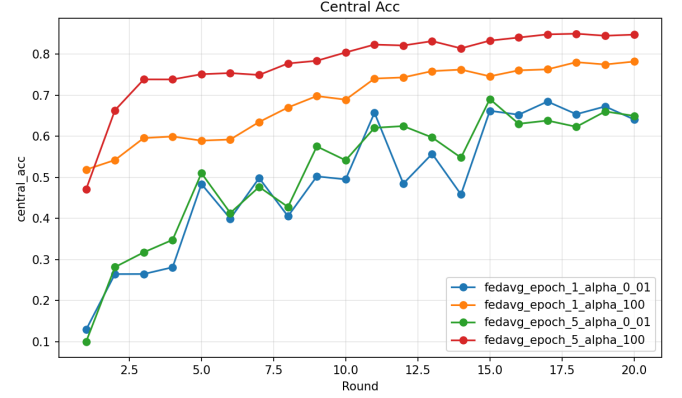


Fig. 6: Central test accuracy for local epochs $\in \{1, 5\}$

E. FedSGD vs FedAvg+SGD vs FedAvg+ADAM

1) Expected Behavior

FedAvg+SGD is the established baseline and should perform reliably across both heterogeneity settings. FedAvg+Adam should converge faster per round than SGD due to per-parameter adaptive learning rates, particularly under near-IID data ($\alpha = 100$) where gradient statistics are stable enough for Adam's momentum and variance estimates to be meaningful. FedSGD takes exactly one mini-batch gradient step per client per round, eliminating local drift entirely; it should converge more slowly than FedAvg (requiring more rounds for equivalent accuracy).

TABLE VI: Round 10 central accuracy for each algorithm.

Algorithm	Central acc ($\alpha = 0.01$)	Central acc ($\alpha = 100$)
FedAvg + SGD	51.9%	61.1%
FedAvg + Adam	31.7%	68.1%
FedSGD	19.6%	26.9%

2) Observation and Deviations

- **FedSGD - very slow convergence.** FedSGD takes one mini-batch gradient, due to tiny updates the global model barely moves. A higher learning rate of 0.1-0.5 would be needed for FedSGD to produce meaningful steps. In FedSGD the effective per-round update magnitude is proportional to $\eta \times \|batch\|/n$, which at $\eta=0.01$ and batch size 64 is too small to produce meaningful weight changes over 10 rounds. But it provides stability in learning and structurally similar to distributed SGD. (refer Figure 8)
- **FedAvg+Adam underperforms SGD at $\alpha = 0.01$.** At $\alpha = 0.01$ each client trains on few dominant classes.

Adam’s adaptive scaling amplifies these already-strong per-class gradients, producing larger and more conflicting weight updates than SGD, which destabilizes aggregation further. Non-IID conditions induces larger client drift with Adam. (refer Figure 7)

- **FedAvg+Adam outperforms SGD at $\alpha = 100$.** The result reverses under near-IID data With homogeneous client data, gradient magnitudes are consistent across clients and across rounds. The conflicting-gradient problem that affects Adam at $\alpha = 0.01$ is absent here, so the adaptive scaling benefit is realized without the destabilization cost.
- Federated evaluation on local validation sets is biased by local data distribution and gives misleading signals appearing artificially high for non-IID settings and near-random for IID settings (refer Figure 9). Hence, a central level evaluation provides a fair assessment.

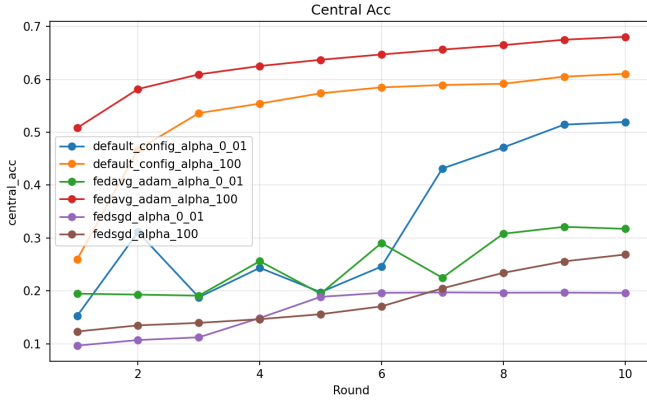


Fig. 7: Central test accuracy for fedavg+sgd, fedavg+adam and fedsgd

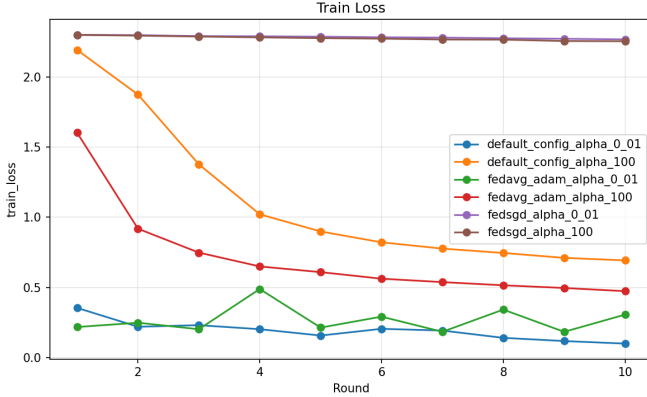


Fig. 8: training loss for fedavg+sgd, fedavg+adam and fedsgd

V. CONCLUSION

Data heterogeneity (alpha) is the dominant factor. Low alpha creates client drift, local models specialize on their dominant classes and produce conflicting gradients. The result is a large

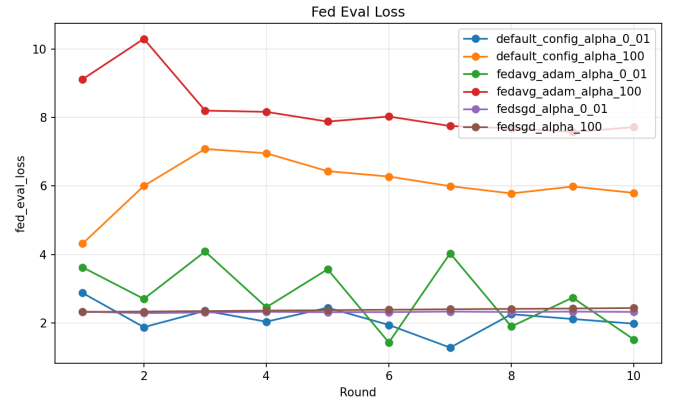


Fig. 9: Federal eval accuracy for fedavg+sgd, fedavg+adam and fedsgd

gap between local training accuracy (high) and global test accuracy (low). High alpha converges more slowly per round but largely stable.

Number of clients primarily affects convergence speed and stability, not final accuracy. Fewer clients means more data per client and faster early convergence. More clients increases round-to-round oscillation under non-IID data because the sampled subset composition varies more. Under moderate heterogeneity clients fail to converge to comparable accuracy within 10 rounds due to conflicting updates, more rounds or a robust aggregation rule would be needed. It adds to the inference heterogeneity is the bigger bottleneck.

Number of local epochs Larger the number of local epochs, greater the effective client drift. Although in IID conditions, higher local epochs converge significantly faster. The trade-off between communication cost and stable learning towards global optimum can be controlled with this hyperparameter.

Sampling fraction is the most operationally critical parameter. At $f=0.1$ with 50 clients and $\alpha=0.1$, only 5 clients participate per round. It acts as trade-off between number of communication cost and stable learning. With higher ratio the model generalizes better but has high communication costs.

Algorithm comparison shows FedAvg+Adam converges faster in early rounds but reaches similar final accuracy to FedAvg+SGD. FedSGD requires many more rounds to match FedAvg because a single batch step per round accumulates knowledge slowly. FedAvg + Adam performs better than FedAvg + SGD as the data become more IID, as large gradient shifts due to momentum leads to client drift for non-IID data.

Sensitive to hyperparameters All experiments are highly sensitive to hyperparameters such as local epochs, number of rounds and learning rate. Future experiments can explore different values to ensure robust results.